
ME 201: Manufacturing Technology - I

Course Code	ME 201
Course Name	Manufacturing Technology - I
L-T-P-C	2-0-2-4
Prerequisites	ME 102
Approval	5th Senate (Oct 2013)

Objective: To impart knowledge about the process principles, equipment, and applications of different forming processes, machining operations, and grinding processes.

Course Contents

- **Introduction:** Classification of different manufacturing processes, application areas and limitations, selection of a manufacturing process.
- **Press Working of Sheet Metal:** Types of presses, drives and feed mechanisms; Operations: Shearing, bending, spinning, embossing, blanking, coining and deep drawing; Die materials, stock layout, compound and progressive dies and punches, construction details of die set, auxiliary equipment, safety devices.
- **Machine Tools and Operations:** Classification of machining processes and machine tools, cutting tool materials, different types of cutting tools, nomenclature of single point and multi point cutting tools, concept of cutting speed, feed and depth of cut, use of coolants, constructional details including accessories and attachment, operations, setting and tooling for capstan and turret lathes, drilling, boring and broaching machines, milling operations.
- **Grinding:** Operations and applications of surface, cylindrical and centreless grinding processes, dressing, truing and balancing of grinding wheels, grading and selection of grinding wheels.

Suggested Books

1. DeGarmo, J. T. Black, Ronald A. Kohser, **Materials and Processes in Manufacturing**, PHI, 1997.
2. Serope Kalpakjian, and Steve R. Schmid, **Manufacturing Engineering and Technology**, Pearson, 2000
3. Groover, M. P., **Fundamentals of Modern Manufacturing**, John Wiley and Sons, 2002.
4. Lindberg, R. A., **Processes and Materials of Manufacture**, PHI, 1990.
5. Rao, P. N., **Manufacturing Technology**, Vol - 2, TMH.